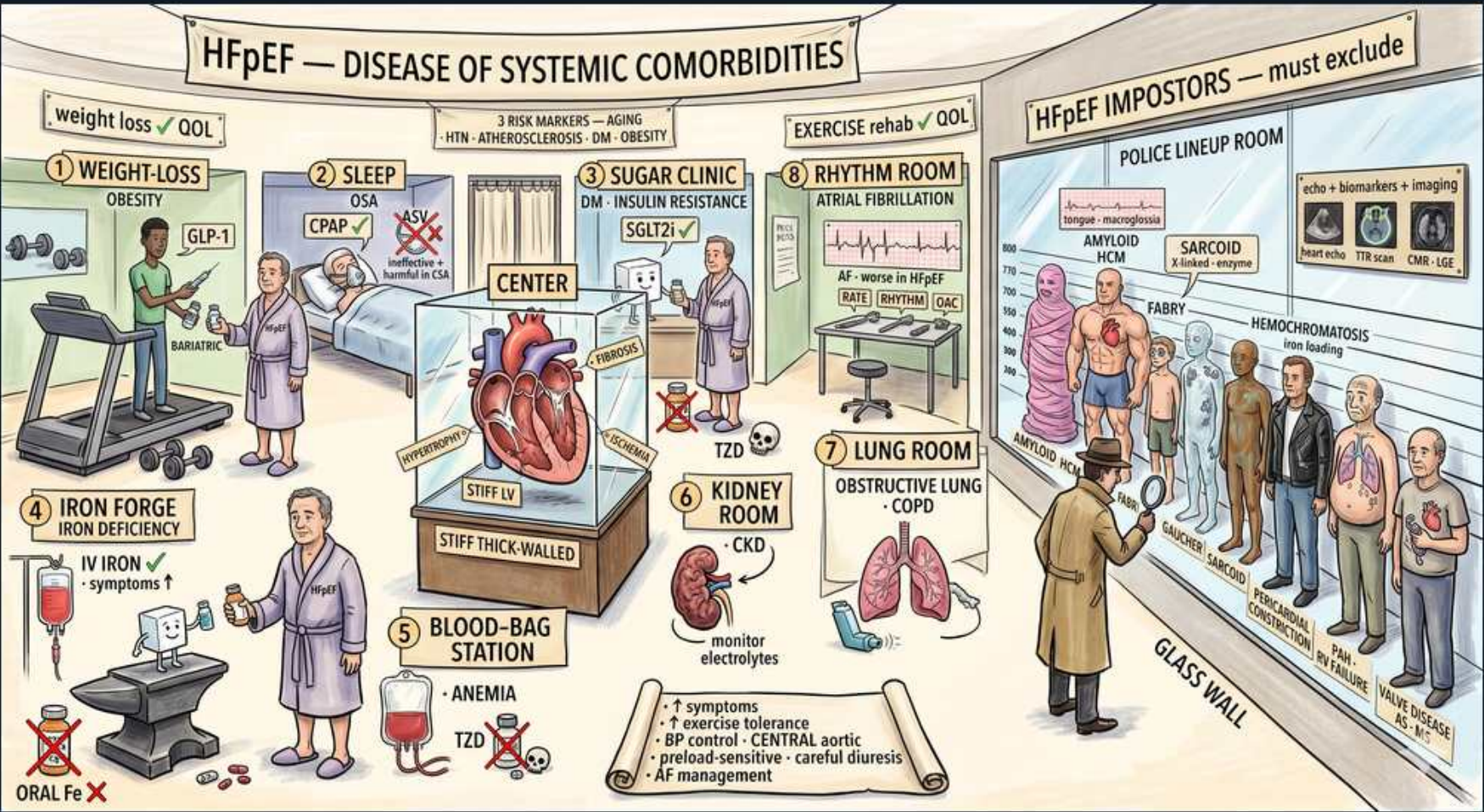


Heart Failure — SCENE 9 — HFpEF: The Crumbling Apothecary



1. HFpEF = disease of systemic comorbidities

WHAT YOU SEE

Banner 'HFpEF — DISEASE OF SYSTEMIC COMORBIDITIES'

WHAT IT ENCODES

HFpEF (HF with preserved EF, LVEF ≥50%) is increasingly framed as a systemic, inflammatory, microvascular syndrome driven by comorbidities — HTN, obesity, diabetes, CKD, AF, OSA, aging — rather than a primary contractile failure. Management therefore targets the comorbidities and congestion; only SGLT2i have robust outcome data. It accounts for roughly half of all HF and predominates in elderly women.

BOARD TRAP

Trap: managing HFpEF with the HFrEF mortality quartet (high-dose ACEi/ARB, beta-blocker, MRA) expecting a survival benefit. WRONG — RAAS inhibitors and beta-blockers have NOT shown clear mortality benefit in HFpEF (CHARM-Preserved, PEP-CHF, TOPCAT largely neutral). Discriminator: in HFpEF, treat comorbidities + congestion and add an SGLT2i; don't transplant the HFrEF regimen wholesale.

2. Stiff thick-walled heart (center)

WHAT YOU SEE

Central thick-walled 'STIFF LV' heart

WHAT IT ENCODES

The HFpEF heart is a stiff, non-compliant, often concentrically hypertrophied LV with impaired relaxation/diastolic dysfunction; EF is preserved (≥50%) but elevated filling pressures cause exertional dyspnea and congestion. Echo shows LVH, left atrial enlargement, elevated E/e' (>14), and elevated estimated PASP; an S4 is the classic auscultatory finding. Natriuretic peptides may be lower than in HFrEF (especially with obesity).

BOARD TRAP

Trap: a stem describing a dilated, thin-walled, hypocontractile ventricle is labeled HFpEF. WRONG — that morphology is HFrEF (eccentric remodeling, low EF). Discriminator: HFpEF = THICK, STIFF, normal-sized cavity with preserved EF and diastolic dysfunction; HFrEF = dilated, thin, reduced EF. Note BNP can be deceptively normal/low in obese HFpEF, so a normal BNP does not exclude it.

3. 1. Weight-loss room (obesity)

WHAT YOU SEE

Treadmill/weights '1 WEIGHT-LOSS / OBESITY', GLP-1, bariatric

WHAT IT ENCODES

Obesity is a major driver of the HFpEF phenotype (epicardial fat, plasma volume expansion, inflammation). Intentional weight loss — caloric restriction, exercise, GLP-1 receptor agonists (semaglutide in STEP-HFpEF improved symptoms/KCCQ and weight), and bariatric surgery — improves symptoms, exercise capacity, and QOL. Combined with exercise training, weight loss is among the few interventions that improve functional status in obese HFpEF.

BOARD TRAP

Trap: dismissing obesity as merely a risk factor and focusing only on diuretics for the obese HFpEF patient. WRONG — weight loss (GLP-1 RA/lifestyle/bariatric) is an evidence-based, symptom-improving therapy in obese HFpEF (STEP-HFpEF). Discriminator: address the obesity phenotype directly; it's disease-modifying for symptoms, not just a background factor.

4. 2. Sleep room (OSA → CPAP)

WHAT YOU SEE

'2 SLEEP / OSA' bed with CPAP □, ASV □

WHAT IT ENCODES

Obstructive sleep apnea is common in HFpEF and worsens nocturnal hypertension, hypoxemia, and AF; treat OSA with CPAP, which improves symptoms and BP. Adaptive servo-ventilation (ASV) is CONTRAINDICATED in HF with CENTRAL sleep apnea and reduced EF — SERVE-HF showed ASV increased cardiovascular mortality in HFrEF with predominant central apnea.

BOARD TRAP

Trap: prescribing ASV for central sleep apnea in a HF patient (especially with reduced EF). WRONG — ASV increased mortality in HFrEF + central apnea (SERVE-HF) and is contraindicated. Discriminator: CPAP is appropriate for OBSTRUCTIVE sleep apnea; ASV is not a safe substitute and is harmful in central apnea with low EF.

5. 3. Sugar clinic (DM, SGLT2i)

WHAT YOU SEE

'3 SUGAR CLINIC / DM, insulin resistance', SGLT2i □

WHAT IT ENCODES

SGLT2 inhibitors (empagliflozin — EMPEROR-Preserved; dapagliflozin — DELIVER) are the FIRST therapy with robust outcome benefit in HFpEF, reducing the composite of CV death/HF hospitalization (driven mainly by fewer hospitalizations) across the EF spectrum and REGARDLESS of diabetes. They are now a guideline-recommended cornerstone of HFpEF therapy; manage coexisting diabetes/insulin resistance as well.

BOARD TRAP

Trap: 'SGLT2i are useless in HFpEF — they were only proven in HFrEF/diabetes.' WRONG — EMPEROR-Preserved and DELIVER proved benefit in HFpEF independent of diabetes. Discriminator: SGLT2i are the key (and essentially only) outcome-modifying drug class in HFpEF; offer to all eligible HFpEF patients, diabetic or not.

6. 3 risk markers: HTN, atherosclerosis, aging, DM, obesity

WHAT YOU SEE

'3 RISK MARKERS — AGING, HTN, ATHEROSCLEROSIS, DM, OBESITY'

WHAT IT ENCODES

The classic HFpEF substrate: advanced AGING, chronic HYPERTENSION (the most important modifiable driver → LVH/diastolic dysfunction), atherosclerosis/CAD, diabetes/insulin resistance, and obesity — clustering most often in elderly women. The H2FPEF and HFA-PEFF scores formalize these (Heavy/BMI, Hypertension ≥2 drugs, AF, Pulmonary HTN, Elder >60, Filling pressure) to estimate probability when the diagnosis is uncertain.

BOARD TRAP

Trap: an elderly hypertensive obese woman with AF and exertional dyspnea but 'normal EF' is told she does not have HF. WRONG — that exact phenotype IS classic HFpEF; use the H2FPEF/HFA-PEFF score and exercise/invasive hemodynamics if needed. Discriminator: preserved EF does NOT rule out HF; the comorbidity cluster + high filling pressures clinches HFpEF.

7. Rhythm room: AF management

WHAT YOU SEE

'6 RHYTHM ROOM / ATRIAL FIBRILLATION — AF worse in HFpEF'

WHAT IT ENCODES

Atrial fibrillation is extremely common in and poorly tolerated by HFpEF patients: the stiff LV is highly dependent on the ATRIAL KICK for filling, so loss of organized atrial contraction and rapid rates sharply raise filling pressures and trigger decompensation. Manage with rate or rhythm control and anticoagulate per CHA2DS2-VASc (HF scores a point).

BOARD TRAP

Trap: treating new AF in a HFpEF patient as a benign incidental finding. WRONG — AF often precipitates acute decompensation in HFpEF because of dependence on the atrial kick. Discriminator: aggressively control rate/rhythm and anticoagulate; restoring sinus rhythm may meaningfully improve symptoms in HFpEF more than in many HFrEF patients.

8. Lung room: COPD

WHAT YOU SEE

'LUNG ROOM — obstructive lung / COPD'

WHAT IT ENCODES

COPD frequently coexists with HFpEF and is a major confounder for the symptom of dyspnea; the two share risk factors (age, smoking) and can each cause exertional breathlessness and exam overlap. Optimize obstructive lung disease, and remember cardioselective beta-blockers (used for other indications) are generally safe in COPD; BNP/echo help disentangle the cause of dyspnea.

BOARD TRAP

Trap: attributing all of a HFpEF patient's dyspnea to COPD (or vice versa) and withholding HF therapy. WRONG — they commonly coexist; missing the cardiac contribution leaves congestion untreated. Discriminator: use BNP/NT-proBNP, echo (E/e', LA size), and response to diuresis to apportion dyspnea; don't reflexively blame the lungs in an elderly hypertensive patient.

9. Kidney room: CKD, monitor electrolytes

WHAT YOU SEE

'KIDNEY ROOM — CKD, monitor electrolytes'

WHAT IT ENCODES

Chronic kidney disease and the cardiorenal syndrome are common in HFpEF and limit diuretic responsiveness while raising hyperkalemia risk with MRAs/RAAS agents. Monitor electrolytes and renal function closely when diuresing or adding spironolactone; worsening creatinine during decongestion is often acceptable if the patient is becoming euvolemic.

BOARD TRAP

Trap: stopping decongestive diuresis the moment creatinine rises slightly in a congested HFpEF patient. WRONG — a modest rise in creatinine during effective decongestion (relief of renal venous congestion) is often acceptable and not true AKI. Discriminator: judge by volume status/perfusion, not creatinine alone; but DO monitor K+ when combining diuretics with MRA/RAAS in CKD.

10. 4. Iron forge (iron deficiency)

WHAT YOU SEE

'4 IRON FORGE / IRON DEFICIENCY', IV iron if symptoms, oral Fe ☐

WHAT IT ENCODES

Iron deficiency (ferritin <100, or 100–299 with TSAT <20%), even without anemia, contributes to fatigue and reduced exercise tolerance in HF. As in HFrEF, IV iron (ferric carboxymaltose) is preferred to improve symptoms/functional capacity; oral iron is poorly absorbed and ineffective in HF (IRONOUT-HF).

BOARD TRAP

Trap: correcting iron deficiency in a symptomatic HFpEF patient with oral ferrous sulfate. WRONG — oral iron is ineffective in HF; use IV iron. Discriminator: screen iron studies in symptomatic HF patients (treat even non-anemic iron deficiency) and choose the INTRAVENOUS route for repletion.

11. 5. Blood-bag station (anemia)

WHAT YOU SEE

'5 BLOOD-BAG STATION / ANEMIA'

WHAT IT ENCODES

Anemia is a frequent comorbidity in HFpEF (from CKD, iron deficiency, chronic inflammation) and worsens dyspnea, fatigue, and outcomes. Work up and treat the underlying cause (iron deficiency, CKD, B12/folate). Avoid liberal transfusion and routine ESAs purely for HF — the RED-HF trial of darbepoetin showed no benefit and increased thromboembolic events.

BOARD TRAP

Trap: giving erythropoiesis-stimulating agents or liberal transfusions to a HF patient just to normalize hemoglobin. WRONG — ESAs did not improve outcomes and raised thromboembolic risk in HF (RED-HF). Discriminator: identify and treat the CAUSE of anemia (often iron deficiency → IV iron); don't reflexively use ESAs/transfusion for the HF itself.

12. BP control / careful diuresis

WHAT YOU SEE

Scroll '↓symptoms, BP control - CENTRAL aortic, preload-sensitive careful diuresis, AF mgmt'

WHAT IT ENCODES

Core HFpEF therapy: control blood pressure (HTN is the leading driver; targeting central aortic load reduces LVH/diastolic dysfunction), relieve congestion with the LOWEST effective dose of loop diuretic, and manage AF. The HFpEF ventricle is steeply PRELOAD-SENSITIVE (operates on a stiff pressure-volume curve), so over-diuresis sharply drops stroke volume → hypotension, prerenal azotemia, and fatigue.

BOARD TRAP

Trap: 'diurese HFpEF aggressively like HFrEF to get them dry.' WRONG — the stiff, preload-sensitive HFpEF ventricle drops cardiac output precipitously with over-diuresis. Discriminator: titrate diuretics carefully to euvolemia (not bone-dry); prioritize sustained BP control as the disease-modifying lever, and watch for hypotension/azotemia from excessive preload reduction.

13. HFpEF impostors — must exclude

WHAT YOU SEE

'HFpEF IMPOSTORS — must exclude' police lineup

WHAT IT ENCODES

Before labeling 'garden-variety' HFpEF, exclude specific, treatable diseases that present with preserved EF and thick walls but have targeted therapies — chiefly cardiac amyloidosis, HCM, sarcoidosis, Fabry disease, and hemochromatosis, plus constrictive pericarditis and high-output states. Missing these means missing disease-specific treatment (e.g., tafamidis, enzyme replacement, immunosuppression, chelation/phlebotomy).

BOARD TRAP

Trap: anchoring on 'HFpEF' in an elderly patient with LVH and not pursuing the mimics. WRONG when red flags exist. Discriminator: clues like low-voltage ECG with 'LVH' on echo, bilateral carpal tunnel, neuropathy/proteinuria, or disproportionate wall thickness should trigger a workup for amyloid (the most commonly missed) and other infiltrative mimics.

14. Mimics: amyloid, HCM, sarcoid, Fabry, hemochromatosis

WHAT YOU SEE

Lineup labeled AMYLOID/HCM, SARCOID, FABRY, HEMOCHROMATOSIS with biomarkers/imaging

WHAT IT ENCODES

Key infiltrative/genetic mimics and their tells: cardiac amyloid (low-voltage ECG with thick walls, apical sparing on strain echo, late gadolinium enhancement on MRI, technetium-PYP scan for ATTR, free light chains for AL); HCM (asymmetric septal hypertrophy, dynamic LVOT obstruction); sarcoid (patchy LGE, AV block, FDG-PET); Fabry (X-linked, neuropathy, angiokeratomas, low alpha-galactosidase A); hemochromatosis (high ferritin/TSAT, T2* MRI). Each has a specific therapy.

BOARD TRAP

Trap: treating biopsy-proven cardiac amyloidosis as ordinary HFpEF, including standard diuretic-heavy/RAAS approaches. WRONG — amyloid patients are sensitive to digoxin and CCBs (drug binding → toxicity) and to over-diuresis/RAAS hypotension; ATTR amyloid has specific therapy (tafamidis). Discriminator: identify the mimic (apical sparing, PYP scan, light chains) and treat the underlying disease rather than applying generic HFpEF management.